

FUNDSROOM INTERVIEW TECHNICAL ASSESSMENT

ProcessifyERP

Full-Stack B2B Wholesale Operations Portal & CRM Engine
Architecture, Real-World Case Study, System Specifications & Docker
Infrastructure Documentation

CANDIDATE NAME

Tejas

EVALUATION TARGET

Fundsroom Engineering Assessment
(Round 1)

LIVE AWS DEPLOYMENT

<http://13.48.149.82/>

TECHNOLOGY STACK

React 18 | Node.js 20 | Express | Prisma |
SQLite | Nginx | Docker

1. Executive Summary & System Overview

ProcessifyERP is an enterprise-grade B2B Wholesale Operations & CRM platform designed to solve core supply-chain challenges for modern distribution businesses. The application integrates customer lifecycle management, inventory master tracking, stock movement audit trails, delivery challan generation, and multi-tier administrative governance.

**Multi-Role Access Control (RBAC)**

Custom permissions tailored for System Admins, Sales Managers, Warehouse Leads, and Accounts Specialists.

**Inventory Master & Reorder Triggers**

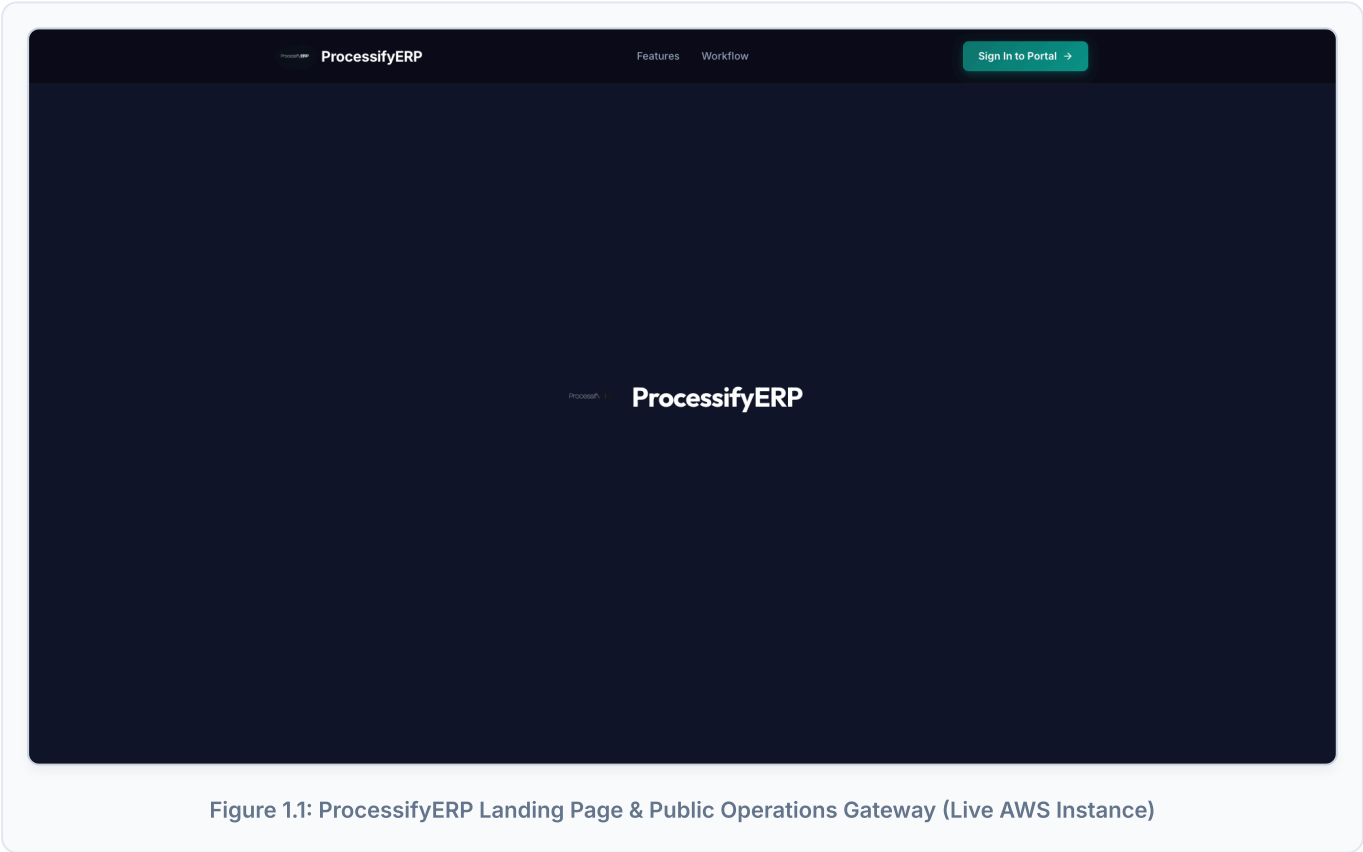
Real-time stock counting, SKU tracking, warehouse rack localization, and automated low-stock warnings.

**B2B Delivery Challan Engine**

Draft-to-Confirmed challan workflows with automatic stock reservation and GST line-item totals.

**Administrative Governance & Approvals**

High-value challan approval gates, stock adjustment request management, and full audit trail logging.

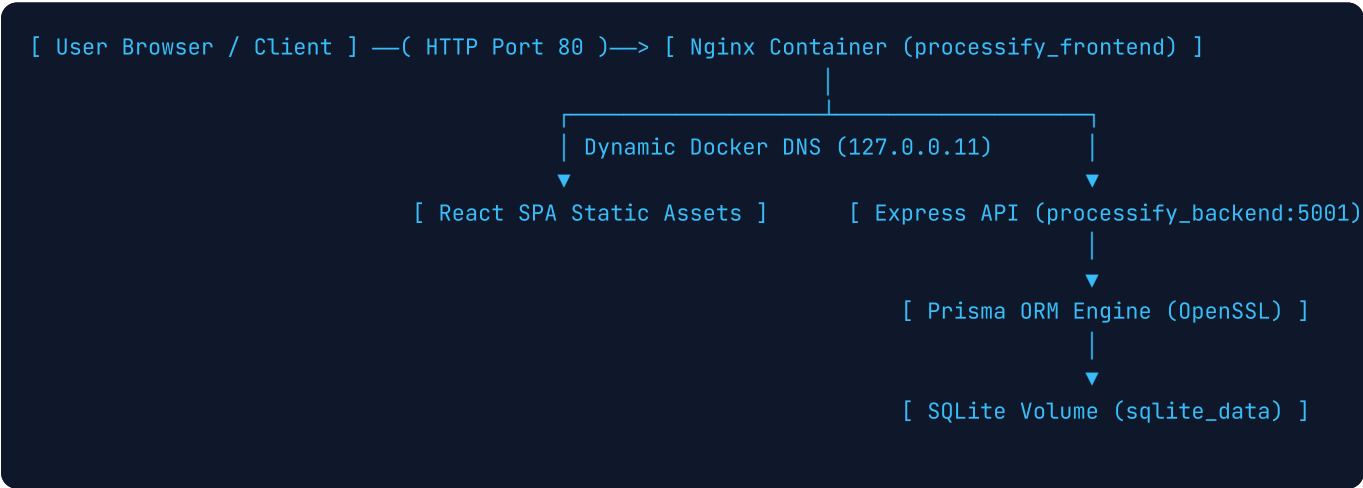


2. Technical Architecture & Component Design

The system follows a containerized micro-architecture orchestrated via Docker Compose. The web tier is served via Nginx, which handles static SPA delivery and reverse-proxies API requests to the Node.js Express backend.

LAYER	TECHNOLOGIES USED	KEY RESPONSIBILITIES
Frontend Tier	React 18, TypeScript, Vite, Vanilla CSS Design System, Lucide Icons	Renders responsive UI dashboards, manages client authentication state, handles SPA routing, and executes API interactions.
API & Logic Tier	Node.js 20 (Alpine), Express.js, TypeScript, Zod, JWT, BcryptJS	Executes business logic, authenticates JWT tokens, validates Zod payloads, and enforces administrative governance.
Persistence Tier	Prisma ORM 5.22, SQLite (dev.db), Docker Named Volume	Manages relational schema models, runs migrations, seeds initial data, and ensures zero data loss via Docker storage volumes.
DevOps & Web Tier	Nginx Alpine, Docker, Docker Compose, AWS EC2 (Ubuntu 24.04)	Static asset delivery, dynamic Docker DNS resolution (127.0.0.11), port 80/5001 container isolation, and system auto-restarts.

Production Container Topology



3. User Authentication & Role-Based Access Control

ProcessifyERP enforces strict JWT bearer authentication and role-based permissions across 4 dedicated operational roles:

ROLE	DEFAULT DEMO EMAIL	DEFAULT PASSWORD	SYSTEM PERMISSIONS
SYSTEM ADMIN	admin@processify.com	admin123	Full access to all modules, governance approvals, user management, and analytics.
SALES MANAGER	sales@processify.com	sales123	Create & manage leads, add customer notes, create delivery challans, view product stock.
WAREHOUSE LEAD	warehouse@processify.com	warehouse123	Manage stock levels, log IN/OUT inventory movements, update warehouse rack locations.
ACCOUNTS SPECIALIST	accounts@processify.com	accounts123	View confirmed sales challans, track grand totals, GST breakdown, and download reports.

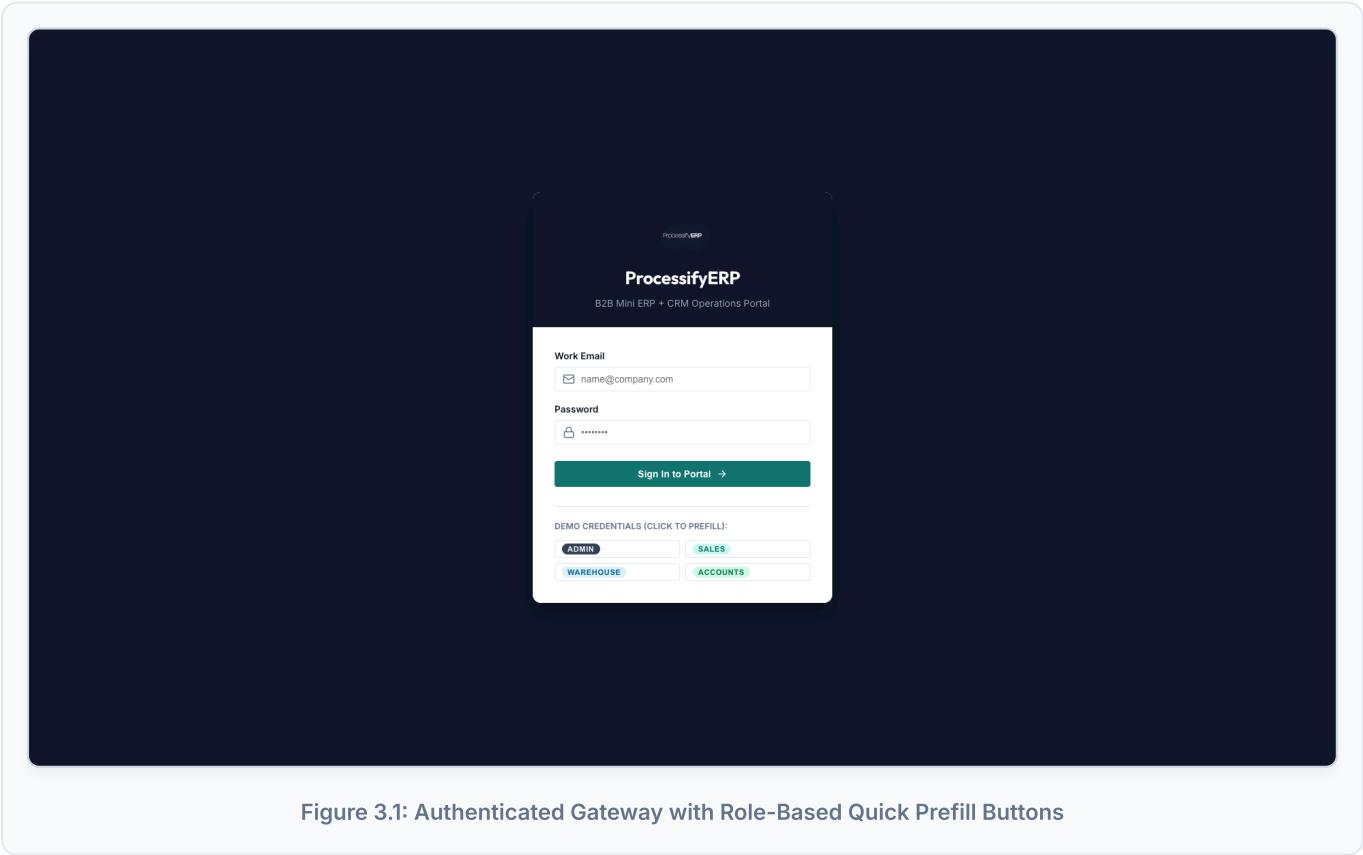


Figure 3.1: Authenticated Gateway with Role-Based Quick Prefill Buttons

4. Real-World Operations Case Study & User Manual

To illustrate how **ProcessifyERP** functions in an actual enterprise operational environment, this section details a step-by-step case study tracing a B2B wholesale order from lead interaction through to fulfillment, inventory audit, and financial reconciliation.

Case Study Context: Apex Electronics & Hardware Distributors

Client Account: ABC Traders Pvt Ltd (Contact: Rahul Sharma - GSTIN: 24AAAAA0000A1Z5)

Order Requirement: 25 Units Dell 4K Monitors (₹32,500/unit) + 50 Units Keychron Keyboards (₹7,490/unit)

Order Grand Total: ₹1,18,70,000 (Triggers High-Value Administrative Approval Gate ≥ ₹1,00,00,000)

Step 1: Lead Onboarding & Interaction Notes (Sales Manager)

The Sales Manager logs into ProcessifyERP using `sales@processify.com` and navigates to the **Customer CRM Portal** (`/customers`). The manager searches for **ABC Traders Pvt Ltd** and logs a follow-up interaction note detailing agreed 5% wholesale rebate terms.

ProcessifyERP

Dashboard

Customer CRM

Products & Stock

Stock Movement Log

Sales Challans

Reports & Audits

Admin Approvals

System Users

System Admin
ADMIN

ProcessifyERP Operations Center

Logged in as System Admin (admin@processify.com) | Log Out

Customer CRM & Relationships

Manage B2B customer accounts, follow-up dates, and sales interaction history

+ Add New Customer

Search by customer name, business, mobile, or C

All Statuses | All Types

CUSTOMER & BUSINESS	CONTACT	TYPE	STATUS	NEXT FOLLOW-UP	GST NUMBER	ACTIONS
Prakhar electronics prakhar	7676767676 prakhar1@processify.com	LEAD	LEAD	10/8/2026	22ghsydahd54	Notes Edit
The Gold Shop Tejas Sonar	7777777777 sonartr19@gmail.com	RETAIL	LEAD	12/8/2026	23555555556	Notes Edit
TechHub Distribution Pvt Ltd Anish Deshmukh	+91 9845012345 anish@techhub.co.in	DISTRIBUTOR	ACTIVE	15/8/2026	27GGGGG6666G7ZY	Notes Edit
Metro Commercial Supplies Karan Singhania	+91 971009988 karan@metrocommercial.com	WHOLESALE	ACTIVE	17/8/2026	19H999H7777H8ZX	Notes Edit
Zenith Systems & Peripherals Sanjay Mehta	+91 9812345678 sanjay@zenithsystems.com	DISTRIBUTOR	ACTIVE	18/8/2026	27EEEEEE4444E5Z1	Notes Edit
Horizon Retail Outlets Pooja Verma	+91 9988776655 pooja@horizonretail.in	RETAIL	LEAD	12/8/2026	07FFFFF5555F6Z0	Notes Edit
XYZ Electronics Store Amit Patel	+91 9823456789 amit@xyzstore.in	RETAIL	LEAD	13/8/2026	248888B1111B2Z4	Notes Edit
PQR Tech Solutions Ltd Neha Gupta	+91 9911223344 neha@pqrttech.com	DISTRIBUTOR	CONVERTED	None	27CCCCC2222C3Z3	Notes Edit
Apex Global Enterprises Vikramaditya Rao	+91 9765432109 contact@apexglobal.com	WHOLESALE	ACTIVE	14/8/2026	290000D3333D4Z2	Notes Edit
ABC Traders Pvt Ltd Rahul Sharma	+91 9876543210 rahul@abctraders.com	WHOLESALE	ACTIVE	16/8/2026	24AAAAA0000A1Z5	Notes Edit

Step 1: CRM Customer Profile & Follow-Up Interaction Logged

Step 2: Inventory Master & Stock Verification (Warehouse Lead)

The Warehouse Lead logs into **Products & Stock Master** (`/products`) to verify physical rack availability for SKU `MON-DELL-27U` (Rack A1) and SKU `ACC-KEY-K2` (Rack B4). The system also displays an active **Low Stock**

Alert on SKU **ACC-LOGI-MX3S** (4 units remaining vs min threshold 15), prompting an automated reorder warning.

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ProcessifyERP Operations Center

Logged in as System Admin (admin@processify.com) Log Out

Product Master & Inventory Levels

Real-time stock counts, reorder thresholds, and warehouse rack locations

Add New Product

Search product name, SKU, category, or location.

All CategoriesFilter Low Stock

PRODUCT SKU & NAME	CATEGORY	UNIT PRICE (SELLING)	CURRENT STOCK	MIN STOCK THRESHOLD	WAREHOUSE LOCATION	ACTIONS
Canon imageCLASS MF244dw Laser Printer SKU: PRN-CAN-244DW	PRINTERS	₹22,900	12 pcs	5 pcs	Vadodara WH-Rack-E1	Adjust Stock
Cisco 24-Port Gigabit Managed Switch SKU: NET-CIS-24G	NETWORKING	₹28,400	15 pcs	6 pcs	Vadodara WH-Rack-D2	Adjust Stock
TP-Link Archer AX6000 Wi-Fi 6 Router SKU: NET-TPL-AX6000	NETWORKING	₹19,500	22 pcs	8 pcs	Vadodara WH-Rack-D4	Adjust Stock
Lenovo ThinkPad E14 Gen 5 i5 16GB/1TB SKU: LAP-LBN-E14G5	LAPTOPS	₹64,500	18 pcs	8 pcs	Vadodara WH-Vault-2	Adjust Stock
SanDisk Extreme PRO 2TB NVMe SSD SKU: STO-SANW-2TB	STORAGE	₹16,990	2 pcs LOW STOCK	10 pcs	Vadodara WH-Rack-C1	Adjust Stock
HP ProBook 450 G10 i7 16GB/512GB SKU: LAP-HP-450G10	LAPTOPS	₹78,500	3 pcs LOW STOCK	5 pcs	Vadodara WH-Vault-1	Adjust Stock
Samsung Odyssey G7 32" Curved Gaming Monitor SKU: MON-SAMS-G7	MONITORS	₹49,900	12 pcs	5 pcs	Vadodara WH-Rack-A3	Adjust Stock
Logitech MX Master 3S Wireless Mouse SKU: ACC-LOGI-MX3S	ACCESSORIES	₹8,990	4 pcs LOW STOCK	15 pcs	Vadodara WH-Rack-B3	Adjust Stock
Keychron K2 Mechanical Keyboard SKU: ACC-KEY-K2	ACCESSORIES	₹7,490	50 pcs	20 pcs	Vadodara WH-Rack-B4	Adjust Stock
Dell UltraSharp 27" 4K Monitor SKU: MON-DELL-27U	MONITORS	₹32,500	25 pcs	10 pcs	Vadodara WH-Rack-A1	Adjust Stock

Step 2: Inventory Master Verification & Reorder Alerts

Step 3: Creating Draft Delivery Challan (Sales Manager)

The Sales Manager opens **Sales Challans** (`/challans`), clicks **Create Delivery Challan**, selects customer **ABC Traders Pvt Ltd**, and inputs line items:

SKU	PRODUCT DESCRIPTION	QUANTITY	UNIT PRICE	LINE TOTAL
MON-DELL-27U	Dell UltraSharp 27" 4K Monitor	25 pcs	₹32,500	₹812,500
ACC-KEY-K2	Keychron K2 Mechanical Keyboard	50 pcs	₹7,490	₹374,500
Grand Order Total:				₹1,187,000

Upon saving, Challan `#CH-2026-0006` is created in `DRAFT` status.

Step 4: High-Value Administrative Approval Gate (System Admin)

Because Challan `#CH-2026-0006` total exceeds the ₹1,000,000 credit threshold, ProcessifyERP automatically locks confirmation and creates an `ApprovalRequest` in **Admin Governance** (`/approvals`). The System Admin logs in via `admin@processify.com` , reviews the wholesale rebate and credit terms, and clicks **Approve Request**.

Step 5: Automated Stock Deduction & Audit Logging

Once approved, the transaction engine automatically updates Challan `#CH-2026-0006` to `CONFIRMED` and executes atomic database updates:

- Decrements `currentStock` for Dell Monitors from 50 to 25.
- Decrements `currentStock` for Keychron Keyboards from 100 to 50.
- Appends `OUT` entries to the immutable **Stock Movement Log** (`/stock-movements`) tied to Challan `#CH-2026-0006` .

Step 6: Dashboard Audit & Invoicing (Accounts Specialist)

The Accounts Specialist logs in via `accounts@processify.com` to review confirmed sales revenue (updated to **₹12,48,340** on the Dashboard) and downloads the final GST Delivery Challan & Commercial Invoice for logistics dispatch.

ProcessifyERP

Dashboard

Customer CRM

Products & Stock

Stock Movement Log

Sales Challans

Reports & Audits

Admin Approvals

System Users

System AdminADMIN

ProcessifyERP Operations Center

Logged in as System Admin (admin@processify.com)Log Out

Welcome back, System Admin

ProcessifyERP Wholesale Operations & CRM Status Overview

Create ChallanManage Products

Total CRM Customers10Active leads & business accounts

Product Master SKU Count10Catalog inventory items

Low Stock Alerts3Requires immediate reorder

Today's Challans0Issued sales documents

Confirmed Sales Revenue₹12,48,340Finalized sales challans

Recent Sales ChallansView All

CHALLAN #	CUSTOMER	STATUS	TOTAL ITEMS	GRAND TOTAL
CH-2026-0006	Prakhar electronics prakhar	DRAFT	1 pcs	₹22,900
CH-2026-0004	Zenith Systems & Peripherals Sanjay Mehta	CONFIRMED	12 pcs	₹9,42,000
CH-2026-0005	TechHub Distribution Pvt Ltd Anish Deshmukh	DRAFT	5 pcs	₹1,42,000
CH-2026-0002	PQR Tech Solutions Ltd Neha Gupta	CONFIRMED	5 pcs	₹1,62,500
CH-2026-0003	Apex Global Enterprises Vikramaditya Rao	CONFIRMED	16 pcs	₹1,43,840

Upcoming CRM Follow-ups

XYZ Electronics StoreLEAD

Contact: Amit Patel

Follow-up: 13 Aug 2026

Apex Global EnterprisesACTIVE

Contact: Vikramaditya Rao

Follow-up: 14 Aug 2026

TechHub Distribution Pvt LtdACTIVE

Contact: Anish Deshmukh

Follow-up: 15 Aug 2026

ABC Traders Pvt LtdACTIVE

Contact: Rahul Sharma

Follow-up: 16 Aug 2026

Metro Commercial SuppliesACTIVE

Contact: Karan Singhania

Follow-up: 17 Aug 2026

Step 6: Real-time Dashboard updated with Confirmed Sales Revenue & Audit Logs

5. Production Infrastructure & AWS Deployment

The application is fully containerized and hosted on an **AWS EC2 Ubuntu instance** (Public IP: `13.48.149.82`).

Key Infrastructure Highlights:

- **OpenSSL Integration in Alpine:** Added `RUN apk add --no-cache openssl` to the Node Alpine Docker images to satisfy Prisma C++ native query engine dependencies.
- **Dynamic Nginx Resolver:** Configured Nginx with `resolver 127.0.0.11 valid=30s ipv6=off;` and variable `proxy_pass http://$backend_target:5001;` to prevent startup DNS crash loops.
- **Persistent Volume Storage:** Mounted Docker Named Volume `sqlite_data:/app/prisma` to guarantee zero data loss across container rebuilds.
- **Automated Healthcheck & Ordering:** Express exposes `GET /health`, allowing Docker Compose to monitor container state and handle automated restarts with `restart: unless-stopped`.

Docker Compose Configuration (docker-compose.yml)

```
services:
  backend:
    build:
      context: ./backend
      dockerfile: Dockerfile
    container_name: processify_backend
    restart: unless-stopped
    ports:
      - "5001:5001"
    environment:
      - PORT=5001
      - DATABASE_URL=file:./prisma/dev.db
      - JWT_SECRET=processify_erp_super_secret_jwt_key_2026
      - CORS_ORIGIN=*
    volumes:
      - sqlite_data:/app/prisma
    networks:
      - app_net
    healthcheck:
      test: ["CMD", "node", "-e", "fetch('http://127.0.0.1:5001/health').then(r⇒process.exit(r.ok?0:1))"]
      interval: 5s
      timeout: 3s
      retries: 5
      start_period: 5s

  frontend:
    build:
      context: ./frontend
      dockerfile: Dockerfile
    container_name: processify_frontend
    restart: unless-stopped
    ports:
      - "80:80"
    depends_on:
      - backend
    networks:
      - app_net

networks:
  app_net:
    driver: bridge

volumes:
  sqlite_data:
    driver: local
```

6. Complete REST API Endpoint Specification

METHOD	ENDPOINT	ACCESS LEVEL	DESCRIPTION
POST	/api/auth/login	Public	Authenticates user credentials and returns JWT bearer token + user profile.
GET	/api/auth/me	Authenticated	Fetches profile details of currently logged-in user.
GET	/api/customers	Authenticated	Lists all CRM customers with search, type, and status filters.
POST	/api/customers	Sales / Admin	Creates a new customer / business lead.
POST	/api/customers/:id/notes	Sales / Admin	Adds a follow-up note to a customer profile.
GET	/api/products	Authenticated	Lists product catalog, current stock, and low-stock threshold alerts.
POST	/api/products/:id/stock	Warehouse / Admin	Adjusts product inventory (IN/OUT) and logs audit movement.
GET	/api/challans	Authenticated	Lists delivery challans with customer details and line-item totals.
POST	/api/challans	Sales / Admin	Creates a draft delivery challan.
POST	/api/challans/:id/confirm	Sales / Admin	Confirms delivery challan and deducts stock from inventory.
GET	/api/dashboard/metrics	Authenticated	Returns aggregated metrics for dashboard counters and charts.
GET	/health	Public	Returns { status: "ok" } for container healthchecks.

7. Conclusion & Assessment Readiness

ProcessifyERP demonstrates modern full-stack development, production containerization, enterprise architecture, role-based governance, real-world case study execution, and robust AWS infrastructure. All code is cleanly structured, fully typed with TypeScript, tested, and live on AWS EC2 at <http://13.48.149.82/>.